Methods

Gaussian elimination

How to solve the standard problems of Lectures 0–6

Most computational problems of the first lectures are solved by one algorithm: write a matrix, reduce it to RREF, and read the answer off the pivots. The problems differ only in *which* matrix you write and *what* you read off. The elimination steps are written out once, in Lecture 0; here they are marked by a single $\xrightarrow{\text{RREF}}$ arrow. Each method ends with the sheet problems it solves.

Choosing the matrix

Vectors go into the *columns*; equations go into the *rows*. A bar appears only when the question names a target. Vectors of any other space are first replaced by their coordinate vectors in one fixed basis; for polynomials and matrices, the standard one (Example 2.15).

Lect.	Question	Matrix	Bar	Read-off
0	Solve $Ax = b$; the null space	equations as rows	$\mid b$	a row $[\,0 \cdots 0 \mid c\,]$, $c \neq 0$: none; else x_p plus one vector per free variable (Section 1)
0	The inverse A^{-1}	A	$\mid I$	left half I: the right half; else no inverse (Section 1)
2	$v \in \operatorname{span}(v_1, \ldots, v_m)$? Coordinates $[v]_{\mathcal{B}}$?	the vectors as columns	$\mid v$	consistent? the solution (Section 2)
2	Independent? A basis of the span?	the vectors as columns, in order	—	the pivot columns (Section 3)
2	$U + W$ and $U \cap W$	bases of U, then W, as columns	—	pivots; relations (Section 4)
3	$\ker T$ and $\operatorname{im} T$, $Tx = Ax$	A	—	free variables; pivot columns of A (Section 5)
4	Matrix of T, basis C of the codomain	C as columns	$\mid Tv_1, \ldots, Tv_n$	the right half (Section 6)
4	Change of basis: $[v]_{\mathcal{B}'}$, A'	P	$\mid [v]_{\mathcal{B}}$; $\mid AP$	the right half (Section 6)
5	Eigenvalues and eigenspaces	$A - \lambda I$ for each root λ of χ_A	—	free variables: a basis of $E(\lambda, A)$ (Section 7)
6	Diagonalization; A^k, e^{tA}	$A - \lambda I$; P	—; $\mid I$	the eigenspace bases make P; P^{-1} (Section 8)

1 Linear systems

Lecture 0

Problem. Solve $Ax = b$: decide whether it has solutions, and write them all. For $b = 0$ this asks for a basis of the null space $\ker A$.

Method

1. Reduce $[A \mid b]$ to RREF. A row $[\,0 \cdots 0 \mid c\,]$ with $c \neq 0$: no solution.
2. Set every free variable to 0 and read off a particular solution x_p.
3. Take $b = 0$. For each free variable, set it to 1 and the other free variables to 0, and read off one vector. These vectors are a basis of $\ker A$.
4. The solution set is x_p plus any linear combination of the basis of $\ker A$.

Example. Solve the system $x_1 - 2x_2 + 2x_3 + x_4 = 2$, $3x_1 - 6x_2 + 5x_3 + 4x_4 = 7$:

$$\left[\begin{array}{cccc|c} 1 & -2 & 2 & 1 & 2 \\ 3 & -6 & 5 & 4 & 7 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cccc|c} \textcircled{1} & -2 & 0 & 3 & 4 \\ 0 & 0 & \textcircled{1} & -1 & -1 \end{array}\right].$$

The free variables are x_2 and x_4, and the pivot variables are $x_1 = 2x_2 - 3x_4 + 4$ and $x_3 = x_4 - 1$. Then:

$$\begin{aligned} x_2 = 0,\ x_4 = 0 &\implies x_p = (4, 0, -1, 0), \\ b = 0,\ x_2 = 1,\ x_4 = 0 &\implies u_1 = (2, 1, 0, 0), \\ b = 0,\ x_2 = 0,\ x_4 = 1 &\implies u_2 = (-3, 0, 1, 1). \end{aligned}$$

The general solution:

$$x = (4, 0, -1, 0) + s\,(2, 1, 0, 0) + t\,(-3, 0, 1, 1),$$

where s and t are arbitrary numbers. ◁

Check. Substitute x_p; multiply A by each null-space vector.

Variant: the inverse. Reduce $[A \mid I]$. If the left half becomes I, the right half is A^{-1}; a zero row on the left means A is not invertible.

Practice. Sheet 0: 0.22, 0.25, 0.26, 0.30, 0.31; 0.29 (split into cases before dividing by an expression in a); 0.23 (null space); 0.24, 0.27 (inverse). Sheet 2: 2.13, 2.16 (the solution set of a homogeneous system).

2 Span membership and coordinates

Lecture 2

Problem. Is v in $\operatorname{span}(v_1, \ldots, v_m)$, and with which coefficients? Coordinates $[v]_{\mathcal{B}}$ ask the same for a basis $\mathcal{B} = (v_1, \ldots, v_m)$.

Method

1. Write $v_1, \ldots, v_m$ into the columns and v after the bar; reduce.
2. A row $[\,0 \cdots 0 \mid c\,]$ with $c \neq 0$: v is not in the span.
3. Otherwise the solution gives the coefficients. For a basis it is unique: it is $[v]_{\mathcal{B}}$.

Example. Is $v = (7, 5, 8)$ in $\operatorname{span}(v_1, v_2)$, where $v_1 = (1, -1, 2)$ and $v_2 = (3, 1, 4)$? We need numbers a and b such that $v = av_1 + bv_2$:

$$\left[\begin{array}{cc|c} 1 & 3 & 7 \\ -1 & 1 & 5 \\ 2 & 4 & 8 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} \textcircled{1} & 0 & -2 \\ 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 \end{array}\right] \implies a = -2,\ b = 3.$$

The last row is zero, so the system is consistent:

$$v = -2v_1 + 3v_2 \quad \text{and} \quad [v]_{\mathcal{B}} = (-2, 3).$$

Suppose instead that $v = (7, 5, 9)$. Then the RREF has the row $[\,0\;0 \mid 1\,]$, and $v \notin \operatorname{span}(v_1, v_2)$. ◁

Practice. Sheet 2: 2.1–2.4 (span); 2.22–2.25 (coordinates).

3 Independence; reducing a list to a basis

Lecture 2

Problem. Is $v_1, \ldots, v_m$ independent? If not, which vectors are redundant, and what are a basis and the dimension of the span?

> **Method**
>
> 1. Write $v_1, \ldots, v_m$ into the columns, in the order of the list; reduce.
> 2. The list is independent iff every column has a pivot.
> 3. A column without a pivot is redundant: its entries are its coefficients in the pivot columns to its left.
> 4. The pivot columns *of the original list* are a basis of the span; the dimension is the number of pivots.

Why step 4 works. Deleting the redundant vectors does not change the span (Lemma 2.10). The same row operations take the remaining columns to distinct standard vectors e_i, so they are independent.

Example. Are $v_1 = (1, 0, 2, 1)$, $v_2 = (2, 1, 3, -1)$, $v_3 = (3, 2, 4, -3)$, $v_4 = (0, 1, 1, 2)$ independent? Find a basis and the dimension of their span.

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 3 & 4 & 1 \\ 1 & -1 & -3 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & -1 & 0 \\ 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies v_3 = -v_1 + 2\,v_2.$$

So $v_1, \ldots, v_4$ are dependent. The vectors v_1, v_2 and v_4 are independent and form a basis of $\operatorname{span}(v_1, v_2, v_3, v_4)$, of dimension 3. ◁

Variants. *Extend* an independent $u_1, \ldots, u_k$ to a basis of $\mathbb{F}^n$: reduce the list $u_1, \ldots, u_k$, $e_1, \ldots, e_n$ and keep the pivot columns. A *square* list with a parameter: the determinant (Example 2.9).

> **Pitfalls**
>
> - Taking the columns of the RREF as the basis: in general they do not even lie in the span.

Practice. Sheet 2: 2.6–2.10, 2.15, 2.19.

4 Sum and intersection of subspaces

Lecture 2

Problem. Given bases $u_1, \ldots, u_k$ of U and $w_1, \ldots, w_l$ of W, find the dimensions and bases of $U + W$ and $U \cap W$.

Method

1. Write $u_1, \dots, u_k$, then $w_1, \dots, w_l$, into the columns; reduce.
2. The pivot columns are a basis of $U + W$ (Section 3). Then $\dim(U \cap W) = k + l - \dim(U + W)$ (Theorem 2.32), the number of columns without a pivot.
3. Each column without a pivot gives a relation. Put its u's on one side and its w's on the other: both sides are one vector, lying in U and in W. These vectors are a basis of $U \cap W$.

Why step 3 works. The u's carry pivots, so every column without one is some w_j, and w_j occurs in no other relation: the vectors are independent, and there are $\dim(U \cap W)$ of them (Proposition 2.24).

Example. Let $u_1 = (1,2,0,1)$, $u_2 = (0,1,1,-1)$, $w_1 = (1,1,0,2)$, $w_2 = (1,2,-1,1)$ and

$$U = \operatorname{span}(u_1, u_2) \quad \text{and} \quad W = \operatorname{span}(w_1, w_2).$$

Find the dimensions and bases of $U+W$ and $U \cap W$. We write u_1, u_2, w_1, w_2 into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 0 & 1 & 0 & -1 \\ 1 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 2 \\ 0 & \textcircled{1} & 0 & -1 \\ 0 & 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies w_2 = 2u_1 - u_2 - w_1.$$

Hence $\dim(U + W) = 3$, $\dim(U \cap W) = 2 + 2 - 3 = 1$, and

$$U + W = \operatorname{span}\big(u_1, u_2, w_1\big), \qquad U \cap W = \operatorname{span}\big(2u_1 - u_2\big) = \operatorname{span}\big((2,3,-1,3)\big),$$

since the vector $2u_1 - u_2 = w_1 + w_2$ lies in both U and W. ◁

Example. Let $u_1 = (1,0,1,2)$, $u_2 = (1,1,0,1)$, $u_3 = (0,1,2,1)$, $w_1 = (2,2,0,2)$, $w_2 = (0,1,1,0)$, $w_3 = (2,1,0,2)$ and

$$U = \operatorname{span}(u_1, u_2, u_3) \quad \text{and} \quad W = \operatorname{span}(w_1, w_2, w_3),$$

both of dimension 3. Find the dimensions and bases of $U + W$ and $U \cap W$. We write the six vectors into the columns of a matrix and reduce it:

$$\begin{bmatrix} 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 2 & 1 & 1 & 2 & 0 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & 2 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 1 \end{bmatrix}.$$

The fourth and sixth columns give $w_1 = 2u_2$ (so w_1 already lies in U) and $w_3 = u_1 + u_2 - u_3 + w_2$. Hence $\dim(U + W) = 4$, that is, $U + W = \operatorname{span}(u_1, u_2, u_3, w_2) = \mathbb{R}^4$, and $\dim(U \cap W) = 3 + 3 - 4 = 2$:

$$U \cap W = \operatorname{span}\big(2u_2,\ u_1 + u_2 - u_3\big) = \operatorname{span}\big((2,2,0,2), (2,0,-1,2)\big),$$

since $2u_2 = w_1$ and $u_1 + u_2 - u_3 = w_3 - w_2$ lie in both U and W. ◁

Pitfalls

- With the vectors in rows, as in Example 2.35, the RREF gives $\dim(U + W)$ but no relations, and so no basis of $U \cap W$.

Practice. Sheet 2: 2.27–2.30.

5 Kernel and image of a matrix map

Lecture 3

Problem. For $T\colon \mathbb{F}^n \to \mathbb{F}^m$, $Tx = Ax$, find bases and dimensions of $\ker T$ and $\operatorname{im} T$; decide whether T is injective or surjective.

> **Method**
>
> 1. Reduce the matrix A. There is no right-hand side, since it stays zero throughout.
> 2. *Kernel:* one vector per free variable, as for the null space in Section 1; $\dim \ker T$ is the number of free variables.
> 3. *Image:* $\operatorname{im} T$ is spanned by the columns of A (Example 3.21), so reduce that list as in Section 3: the pivot columns of A — not of its RREF — are a basis, and $\operatorname{rank} T$ is the number of pivots.

Example. Let $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ be $Tx = Ax$. Find bases and the dimensions of $\ker T$ and $\operatorname{im} T$. We reduce the matrix A:

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 2 & 3 & -5 & 4 \\ -1 & 1 & -5 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 2 & -1 \\ 0 & \textcircled{1} & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies x_1 = -2x_3 + x_4,\ x_2 = 3x_3 - 2x_4.$$

The free variables are x_3 and x_4, and the pivot variables are x_1 and x_2. Then:

$$\begin{aligned} x_3 = 1,\ x_4 = 0 &\implies x = (-2, 3, 1, 0), \\ x_3 = 0,\ x_4 = 1 &\implies x = (1, -2, 0, 1). \end{aligned}$$

Hence

$$\ker T = \operatorname{span}\big((-2, 3, 1, 0), (1, -2, 0, 1)\big), \qquad \operatorname{im} T = \operatorname{span}\big((1, 2, -1), (2, 3, 1)\big).$$

Both have dimension 2: T is neither injective nor surjective. ◁

Check. Multiply A by each kernel vector.

> **Pitfalls**
>
> - The pivot columns of the RREF, $(1, 0, 0)$ and $(0, 1, 0)$, span the plane $x_3 = 0$, which does not contain $(1, 2, -1)$: row operations change the column space.
> - $4 = 2 + 2$ is no check: pivots and free variables always add up to n.

Variant: a map from its values on a basis (Example 3.7). Find the coordinates $a, b, \dots$ of $(x, y, \dots)$ in that basis by Section 2, with $(x, y, \dots)$ after the bar; then $T(x, y, \dots) = a\,Tv_1 + b\,Tv_2 + \cdots$.

Practice. Sheet 3: 3.8, 3.10–3.12, 3.17, 3.22, 3.23, 3.29(a); 3.4 (a map from its values).

6 Several right-hand sides at once

Lecture 4

Problem. Find the matrix of a map in given bases, convert coordinates, or find the matrix $A' = P^{-1}AP$ of an operator in a new basis.

Method

1. Row operations never look at the right-hand side, so systems that share one matrix M are solved together. If M is invertible — as it is when its columns are a basis — $[\,M \mid B\,]$ reduces to $[\,I \mid M^{-1}B\,]$. If the left half does not become I, M is not invertible.
2. Choose M and B from the table below.

Question	M	B	Result
Inverse (Section 1)	A	I	A^{-1}
Matrix of T, basis C of the codomain	C as columns	$Tv_1, \dots, Tv_n$ as columns	$\mathcal{M}(T)$
New coordinates	P	$[v]_{\mathcal{B}}$	$[v]_{\mathcal{B}'}$
New matrix (Theorem 4.26)	P	AP	A'

All columns are coordinate vectors in one fixed basis of the codomain. P holds the new basis vectors as columns, written in the old basis (Definition 4.24). If the old basis $\mathcal{B} = (v_1, \dots, v_n)$ is not the standard one, first find P itself: reduce $[\,v_1 \cdots v_n \mid v_1' \cdots v_n'\,]$ (all vectors in standard coordinates) to $[\,I \mid P\,]$. The other direction needs no reduction: $[v]_{\mathcal{B}} = P\,[v]_{\mathcal{B}'}$.

Example. An operator has the matrix $A = \left(\begin{smallmatrix} 4 & -1 \\ 2 & 1 \end{smallmatrix}\right)$ in the standard basis. Find its matrix A' in the new basis $(1,1), (2,3)$ of $\mathbb{R}^2$. Then $P = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 3 \end{smallmatrix}\right)$, $AP = \left(\begin{smallmatrix} 3 & 5 \\ 3 & 7 \end{smallmatrix}\right)$, and

$$\left[\begin{array}{cc|cc} 1 & 2 & 3 & 5 \\ 1 & 3 & 3 & 7 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|cc} \textcircled{1} & 0 & 3 & 1 \\ 0 & \textcircled{1} & 0 & 2 \end{array}\right] \Longrightarrow A' = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

The right half is the matrix of the operator in the new basis. ◁

Example (A non-standard old basis). Let $\mathcal{B} = \big((1,1), (1,2)\big)$ and $\mathcal{B}' = \big((2,3), (1,1)\big)$. A vector has $[v]_{\mathcal{B}} = (1,2)$, and an operator has $A = \mathcal{M}(T, \mathcal{B}) = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$. Find $[v]_{\mathcal{B}'}$ and $\mathcal{M}(T, \mathcal{B}')$. First we find P:

$$\left[\begin{array}{cc|cc} 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|cc} \textcircled{1} & 0 & 1 & 1 \\ 0 & \textcircled{1} & 1 & 0 \end{array}\right] \Longrightarrow P = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $AP = \left(\begin{smallmatrix} 3 & 1 \\ 7 & 3 \end{smallmatrix}\right)$, and one reduction gives both answers:

$$\left[\begin{array}{cc|c|cc} 1 & 1 & 1 & 3 & 1 \\ 1 & 0 & 2 & 7 & 3 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c|cc} \textcircled{1} & 0 & 2 & 7 & 3 \\ 0 & \textcircled{1} & -1 & -4 & -2 \end{array}\right].$$

So $[v]_{\mathcal{B}'} = (2, -1)$ and $\mathcal{M}(T, \mathcal{B}') = \left(\begin{smallmatrix} 7 & 3 \\ -4 & -2 \end{smallmatrix}\right)$. ◁

Check. $PA' = AP$.

Pitfalls

- Reducing $[\,P \mid A\,]$ gives $P^{-1}A$, not A': multiply by P first.

Practice. Sheet 4: 4.9, 4.11, 4.12 (if the left half does not become I, find a kernel vector by Section 5); 4.16–4.21; 4.10, 4.13 (Section 3, after coordinates).

7 Eigenvalues and eigenspaces

Lecture 5

Problem. Find the eigenvalues of a square matrix A and a basis of each eigenspace $E(\lambda, A) = \ker(A - \lambda I)$.

Method

1. Form the characteristic polynomial $\chi_A(\lambda) = \det(\lambda I - A)$. Expand the determinant along the row or column with the most zeros. For a 2×2 matrix,
$$\chi_A(\lambda) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$
2. Solve $\chi_A(\lambda) = 0$. The roots are the eigenvalues.
3. Check: all the roots, complex ones included, add up to $\operatorname{tr} A$ and multiply to $\det A$ (Proposition 5.8). Count each root as many times as its multiplicity.
4. For each eigenvalue λ, reduce the matrix $A - \lambda I$ and read off one vector per free variable, as for the null space in Section 1. These vectors are a basis of $E(\lambda, A)$.

Example. Find the eigenvalues of the matrix
$$A = \begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$$
and a basis of each eigenspace. We expand the determinant along the first row:
$$\chi_A(\lambda) = \det \begin{pmatrix} \lambda - 3 & 0 & 0 \\ -2 & \lambda - 1 & 2 \\ -2 & 2 & \lambda - 1 \end{pmatrix} = (\lambda - 3)\big((\lambda - 1)^2 - 4\big) = (\lambda - 3)^2(\lambda + 1).$$
So the eigenvalues are 3 (multiplicity 2) and -1. Check:
$$3 + 3 + (-1) = 5 = \operatorname{tr} A \quad \text{and} \quad 3 \cdot 3 \cdot (-1) = -9 = \det A.$$
For $\lambda = 3$ we reduce the matrix $A - 3I$:
$$\begin{bmatrix} 0 & 0 & 0 \\ 2 & -2 & -2 \\ 2 & -2 & -2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies x_1 = x_2 + x_3.$$
The free variables are x_2 and x_3. Then:
$$\begin{aligned} x_2 = 1,\ x_3 = 0 &\implies x = (1, 1, 0), \\ x_2 = 0,\ x_3 = 1 &\implies x = (1, 0, 1). \end{aligned}$$
Hence
$$E(3, A) = \operatorname{span}\big((1, 1, 0), (1, 0, 1)\big).$$
For $\lambda = -1$ we reduce the matrix $A + I$:
$$\begin{bmatrix} 4 & 0 & 0 \\ 2 & 2 & -2 \\ 2 & -2 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 \end{bmatrix} \implies x_1 = 0, \quad x_2 = x_3.$$
So the free variable is x_3. Setting $x_3 = 1$, we read off the basis vector $(0, 1, 1)$ and
$$E(-1, A) = \operatorname{span}\big((0, 1, 1)\big).$$
The double eigenvalue 3 has a two-dimensional eigenspace. ◁

Check. Verify $Av = \lambda v$ for each vector found. If the RREF of $A - \lambda I$ has no free variable, λ is not an eigenvalue: look for the error in χ_A, in its roots or in the reduction.

Pitfalls

- Eigenvalues cannot be read off the diagonal of an RREF: row operations change the eigenvalues. The diagonal holds the eigenvalues only when A itself is triangular.
- A double root does not guarantee two independent eigenvectors: $\dim E(\lambda, A)$ is the number of free variables and can be smaller than the multiplicity (Example 5.11).

Variants. A *triangular* matrix: the eigenvalues are its diagonal entries (Theorem 5.21, Example 5.10), so steps 1–2 are not needed. A *real* matrix, working over $\mathbb{C}$: for a non-real eigenvalue λ, reduce $A - \lambda I$ only. The conjugates of the basis vectors found are a basis of $E(\overline{\lambda}, A)$, since $A\overline{v} = \overline{Av} = \overline{\lambda}\,\overline{v}$. An *operator* on another space: take its matrix in some basis (for polynomials, the standard one). The vectors found are the coordinate vectors of the eigenvectors: turn them back into vectors of that space.

Practice. Sheet 5: 5.1, 5.4, 5.5, 5.8; 5.1(iv), 5.2 (over $\mathbb{C}$); 5.3 (a space of polynomials); 5.13 (a triangular matrix).

8 Diagonalization and functions of a matrix

Lecture 6

Problem. Decide whether a square matrix A is diagonalizable. If it is, find P and D with $A = PDP^{-1}$, and compute A^k or e^{tA}.

Method

1. Find the eigenvalues and a basis of each eigenspace (Section 7).
2. If the bases together hold fewer than n vectors, A is not diagonalizable (Theorem 6.2). If it has n distinct eigenvalues in the field used, it always is (Corollary 6.3).
3. Put the basis vectors into the columns of P, and their eigenvalues, in the same order, on the diagonal of D. Then $A = PDP^{-1}$.
4. $f(A) = P\,f(D)\,P^{-1}$, where f is applied to each diagonal entry of D (Definition 6.17): λ^k for A^k, $e^{\lambda t}$ for e^{tA}. Find P^{-1} by reducing $[\,P \mid I\,]$ (Section 1); for a 2×2 matrix the formula of Lecture 0 is enough.

Example. Diagonalize the matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 2 \end{smallmatrix}\right)$ and compute A^k. Since $\operatorname{tr} A = 3$ and $\det A = -4$,

$$\chi_A(\lambda) = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1),$$

so the eigenvalues are 4 and -1. For $\lambda = 4$ we reduce the matrix $A - 4I$:

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & -\frac{2}{3} \\ 0 & 0 \end{bmatrix} \Longrightarrow\ x_1 = \frac{2}{3}x_2.$$

To avoid a fraction we set $x_2 = 3$ and get the eigenvector $(2, 3)$. For $\lambda = -1$ we reduce the matrix $A + I$:

$$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \textcircled{1} & 1 \\ 0 & 0 \end{bmatrix} \Longrightarrow\ x_1 = -x_2.$$

Setting $x_2 = 1$, we get the eigenvector $(-1, 1)$. The eigenvalues are distinct, so A is diagonalizable:

$$P = \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{5}\begin{pmatrix} 1 & 1 \\ -3 & 2 \end{pmatrix}.$$

Then

$$A^k = PD^kP^{-1} = \frac{1}{5}\begin{pmatrix} 2\cdot 4^k & -(-1)^k \\ 3\cdot 4^k & (-1)^k \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -3 & 2 \end{pmatrix},$$

so

$$A^k = \frac{4^k}{5}\begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} + \frac{(-1)^k}{5}\begin{pmatrix} 3 & -2 \\ -3 & 2 \end{pmatrix}.$$

Replacing 4^k and $(-1)^k$ by e^{4t} and e^{-t} would give e^{tA}. ◁

Check. $AP = PD$ (no inverse needed). The formula for A^k must give I at $k = 0$ and A at $k = 1$.

Pitfalls

- $A = PDP^{-1}$, not $P^{-1}DP$; the order on the diagonal of D must match the order of the columns of P.
- The entries of e^{tA} are not $e^{ta_{ij}}$: f acts on the eigenvalues, not on the entries.

Variants. *The equation* $\dot{x} = Ax, x(0) = x_0$: no P^{-1} is needed. Find the coordinates $c_1, \dots, c_n$ of x_0 in the eigenvector basis by reducing $[\,P \mid x_0\,]$ (Section 2). Then $x(t) = c_1e^{\lambda_1 t}v_1 +$

$\cdots + c_n e^{\lambda_n t} v_n$, where the v_k are the columns of P (Example 6.22). *Non-real* eigenvalues: diagonalize over $\mathbb{C}$; for a real matrix, e^{tA} still comes out real. *The Cayley–Hamilton theorem* $\chi_A(A) = 0$ (Theorem 6.11) gives A^{-1} and high powers without eigenvectors. For a 2×2 matrix it reads $A^2 = (\operatorname{tr} A)A - (\det A)I$: apply this repeatedly until A^k is written as $\alpha A + \beta I$. When $\det A \neq 0$, move the constant term of $\chi_A(A) = 0$ to the other side and factor out A (Corollary 6.12); for a 2×2 matrix, $A^{-1} = \big((\operatorname{tr} A)I - A\big)/\det A$.

Practice. Sheet 6: 6.1(i)–(ii); 6.1(iii) (over $\mathbb{C}$); 6.1(iv)–(v) (not diagonalizable); 6.2 (A^n); 6.10(i)–(ii), 6.7; 6.10(iv) (over $\mathbb{C}$); 6.10(iii) (the equation $\dot{x} = Ax$); 6.4, 6.5 (the Cayley–Hamilton theorem).